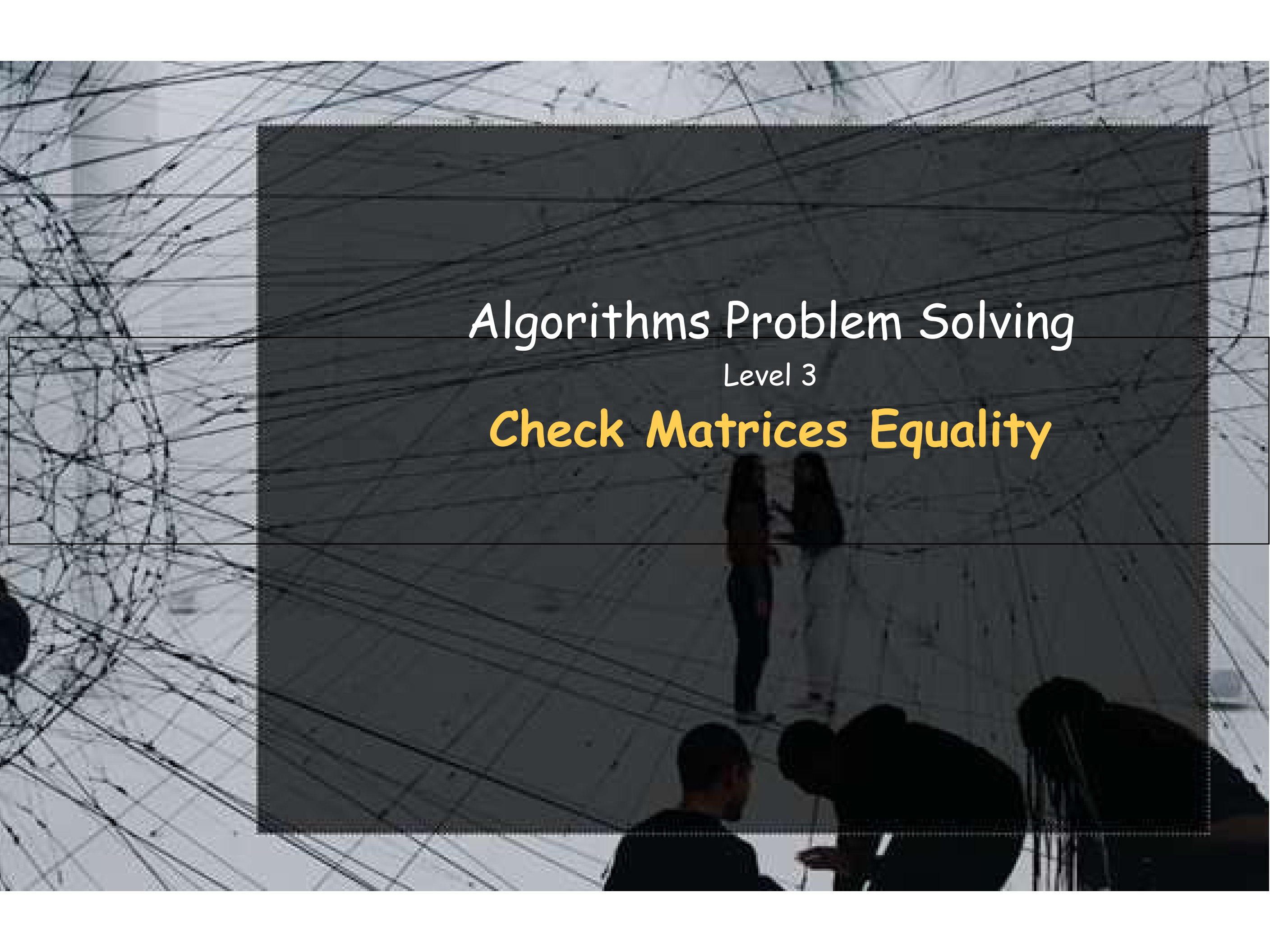


Algorithms Problem Solving

Level 3

Check Matrices Equality



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Definition

Sum-Based Matrix Comparison

 Program generates two 3×3 matrices and compares them by calculating the total sum of all their elements.

 The comparison reduces each matrix to one scalar value and checks whether those two sums are equal.

Explanation

Matrix Sum Calculation

-  Every element in Matrix1 and Matrix2 is visited using nested loops.

-  The values are accumulated into separate sum variables.

Scalar Equality Check

-  After both totals are calculated, the program compares the two sums using a boolean condition.

-  If the sums are equal, the function returns `true`; otherwise, it returns `false`.

Aggregate Comparison Concept

-  This approach compares the overall numerical value of each matrix rather than comparing individual positions.

-  It is therefore an aggregate comparison, not a true structural matrix equality test.

Procedures

1. Create two 3×3 matrices with random numbers.
2. Print both matrices.
3. Calculate each matrix sum using `SumOfMatrix()`.
4. Compare the two sums inside `AreMatricesEqual()`.
5. Display YES if the sums are equal; otherwise display NO.

💡 Example

Matrix1:

6 3 6

4 5 9

3 4 4

Matrix2:

8 9 6

6 8 6

9 3 5

➡ Result

No: matrices are NOT equal.

Note

- ◆ Sum comparison can be useful when only the aggregate value of a matrix matters.
- ◆ The two matrices may contain completely different elements and still produce the same sum.
- ◆ The comparison itself is $O(1)$ after both sums have already been calculated, while calculating the sums is $O(N \times M)$.

⊘ Warning

- ◆ Equal sums do NOT prove that two matrices contain equal elements.
- ◆ Matrices such as $\begin{bmatrix} 1 & 2 \end{bmatrix}$ and $\begin{bmatrix} 2 & 1 \end{bmatrix}$ have the same sum but different element positions.
- ◆ Do not use sum equality when exact matrix identity or structural equality is required.



Comparison

▨ `AreMatricesEqual()` by Sum

- ✓ Compares only the total scalar sum of each matrix.
- ✓ Can be useful when aggregate totals are the actual requirement.
- ✓ May return `true` for completely different matrices with equal sums.

▤ `AreMatricesIdentical()` by Element

- ✓ Compares corresponding elements at the same positions.
- ✓ Detects differences in both values and positions.
- ✓ Provides true content equality when dimensions are also equal.

✦✦ Summary

- ✓ Calculates the total value of every element in each matrix.
- ✓ Reduces each matrix to a single scalar sum.
- ✓ Compares the resulting sums using boolean logic.
- ✓ Demonstrates why aggregate equality is different from element-by-element equality.

Thank you 😊